

13-8-24.

Casting $\rightarrow$ Solidification.

Welding $\rightarrow$ Melting / fusion process.

Forming $\rightarrow$ Plastic deformation.

Cutting $\rightarrow$ shearing

Work material $\rightarrow$ cutting tool $\rightarrow$ M/C tool $\rightarrow$ Product.

Machining condition.

$\rightarrow$ Turning, boring, planing $\rightarrow$ single, continuous.

$\rightarrow$ Drilling, milling, grinding $\rightarrow$ multipoint, intermittent.

Machining process $\rightarrow$ Metal cutting process to generate surface by relative motion b/w workpiece & tool.

29-8-24.

Machining process.

Limitations of casting & forming :

Casting imposes several problems from the point of view of material properties and accuracy whereas, forming becomes impractical when the job is either very large (very large forming process require huge machine) or the material is not suitable for the forming operation. Apart from these, the geometric complexities of the final job such that these methods are of no use.

BBC material.

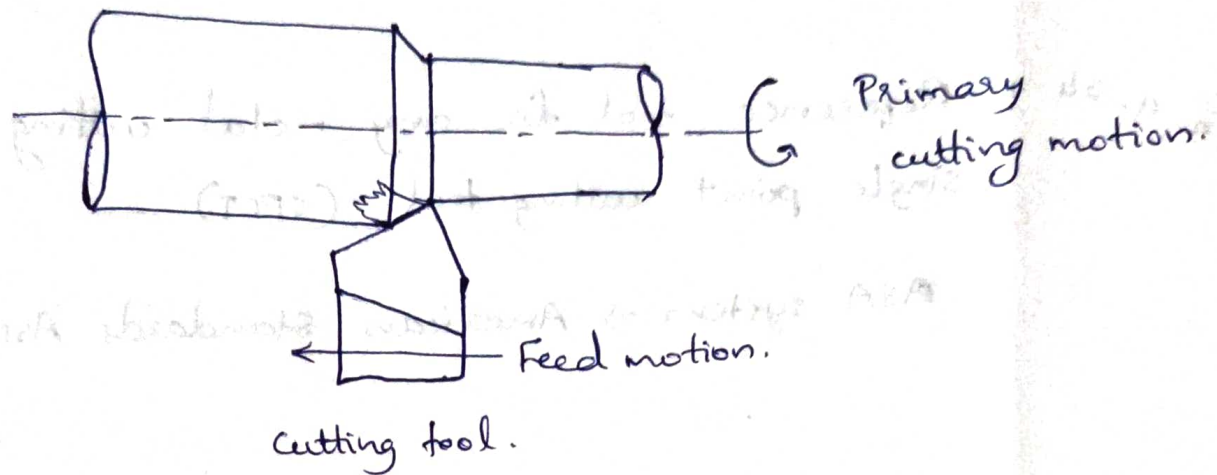
These problems can be overcome if the method used is such that the desired shape, size & finish are obtained through the removal of excess material in the form of small chips. In most cases, a large scale removal is not possible either for geometric reasons or for the size involved. This process is termed as machining [it is perhaps the most versatile manufacturing process]. The body which removes excess material through a direct mechanical contact is called a cutting tool & the machine which provides necessary motion b/w the work piece & tool is commonly known as machine tool. (Eg. lathe).

Drill bit $\Rightarrow$ cutting tool.

Drilling machine $\Rightarrow$ Machine tool.

Since the removal of material takes place only in the form of small chips, the machining of a finite area requires a continuous feeding of the uncut portion at a suitable rate. Relative motion b/w tool & work responsible for ----
& that responsible for feeding gradually is called secondary or feed motion.

Turning.



Geometry of cutting tool mainly implies the tool angle

- Rake angle: angle b/w rake surface & reference plane
- Clearance angle (α): angle b/w flank surface from finished surface.

Avoids rubbing of tool with machined surface which causes energy loss & damages both tool & work piece.

clearance angle = $3^\circ - 15^\circ$

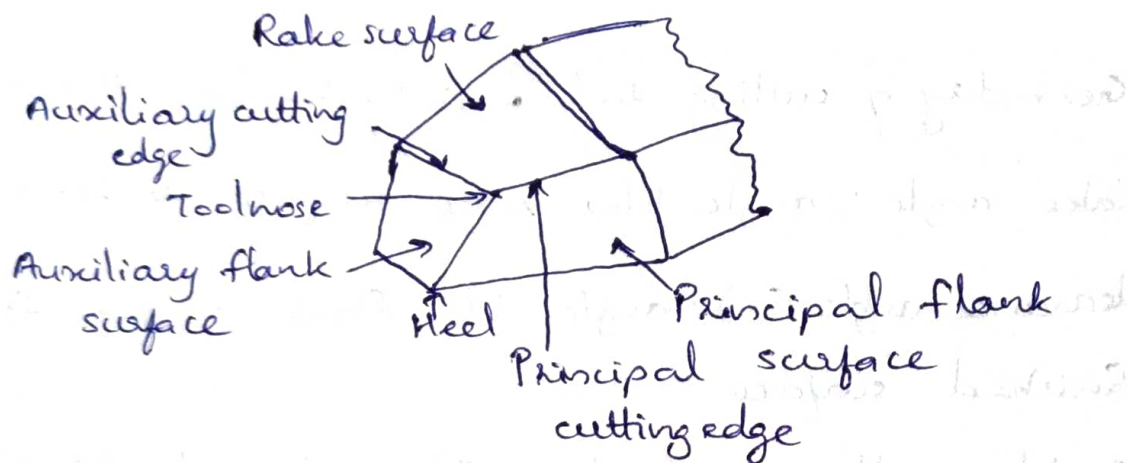
Single point cutting tool has 6 inches. angles. & 1 nose angle.

- Rake angle is provided for ease of chip flow & overall machining. It may be +ve, -ve or 0.
 - +ve rake helps to reduce cutting force & thus ~~cutting power~~ machining cutting power requirement.
 - -ve rake increase edge-strength & life of tool.

- O sake simplify design & manufacture of form tools.

24. → Reference tool for any metal cutting operation is single point cutting tool (SPCT)

ASA system → American Standards Association.



→ Planes of reference & the coordinates used in ASA system:

π_R = Reference plane.

π_x = Machine longitudinal plane.

π_y = Machine traverse plane.

x_m , y_m , z_m are in the direction of longitudinal feed, cross feed & cutting velocity respectively.

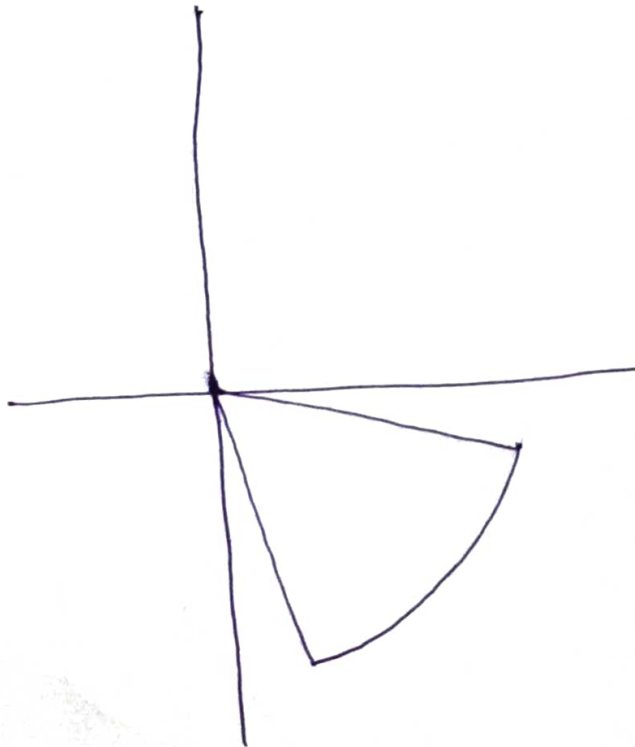
2 rake angle, 2 clearance angle, 2 cutting edge angles are present (total 6) in a SPCT.

$2\psi = \alpha - 2\phi$

For ductile material, we'll get continuous chips & discontinuous chips for non-ductile material.

$a_1 \Rightarrow$ uncut thickness.

$a_2 > a_1 \quad \therefore a_2$ expands.



Chip thickness, a_2

From fig;

$$AC = a_2 = OA \cos(\beta - \phi) \quad \& \quad AB = a_1 = OA \sin \beta$$

$$a_2 / a_1 = r_e = \cos(\beta - \phi) / \sin \beta$$

$$\tan \beta = \cos \phi / r_e - \sin \phi$$

Back rake = γ_y

Side rake = γ_x

Back clearance angle = α_y

Side clearance angle = α_x

Side cutting edge = ψ_s

End cutting edge = ψ_e

Nose radius = r .

Types of metal cutting process

→ Orthogonal cutting process.

Continuous chip flow along orthogonal plane.
Practically it may not flow along it due to inclination angle, λ .

When, $\lambda = 0^\circ$, → chip flow along orthogonal plane; $P_c = 0$

$\lambda \neq 0^\circ$, → chip flow is deviated from π_0 & $P_c = \lambda$.

where P_c = chip flow deviation.

→ Oblique cutting process.

chip flow deviates from orthogonal plane, $P_c \neq 0^\circ$.

chip formation.

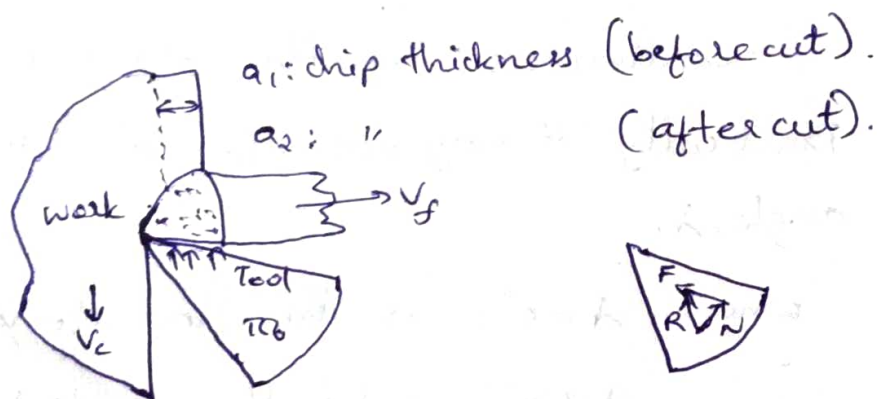
- Compression
- Deformation
- Break.

Machining is a process of gradually removing excess material from performed blanks in the form of chips.

Form of chips indicates:

- Nature & behaviour of work material under machining condition.
- Specific energy requirement.
- Nature & degree of interaction at chip-tool interface.

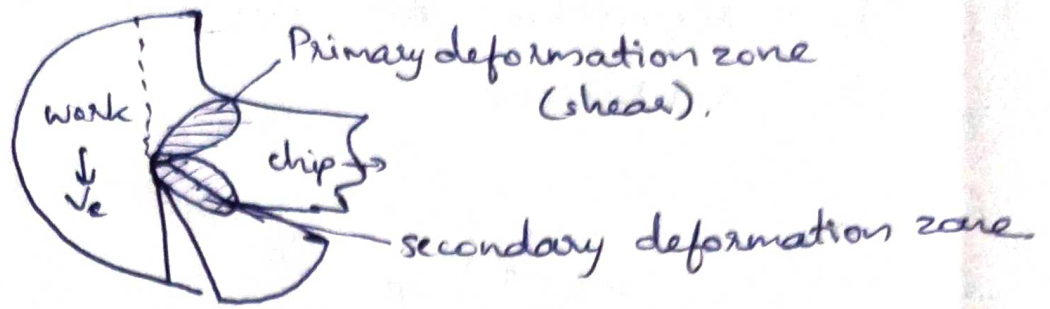
Mechanism of chip formation in ductile materials.



During continuous machining, uncut layer of work material just ahead of cutting tool is subjected to almost all sided compression.

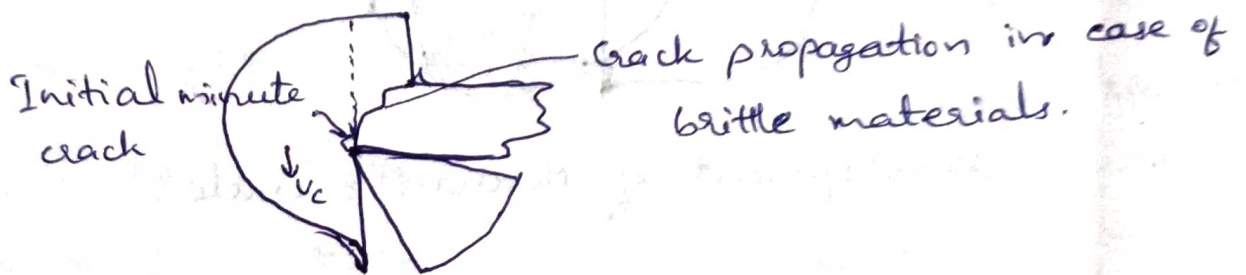
Pattern & extend of total deformation depends on:

- Work material
- Tool material & geometry
- Machining speed.
- Cutting fluid application.



Mechanisms of chip formation in brittle materials.

- 1). Yielding $\rightarrow$ generally for ductile materials.
- 2). Brittle fracture $\rightarrow$ " brittle "



chip thickness $>$ uncut chip thickness due to:

- $\rightarrow$ compression of chip ahead of tool.
- $\rightarrow$ Frictional resistance to chip flow
- $\rightarrow$ lamellae sliding according to piispanen.

P_x = Feed direction. (least imp).

P_y = cross feed. (transverse direction).

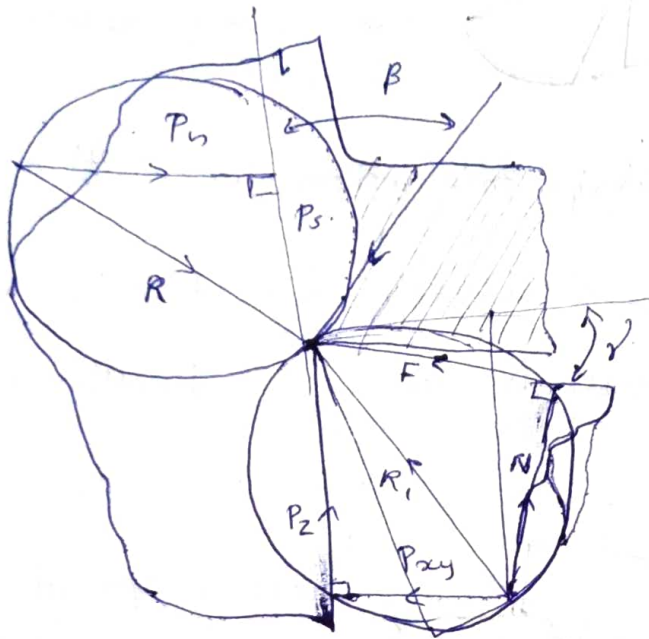
R = Resultant force.

$$R = P_z + P_{xy}.$$

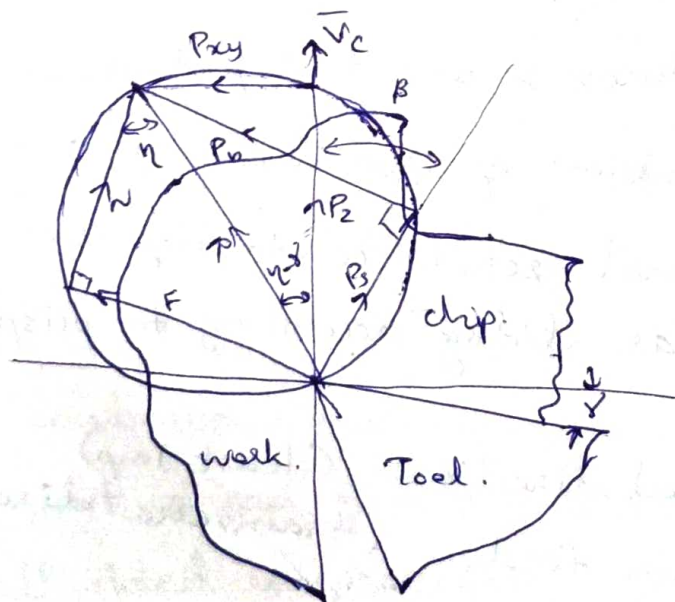
P_z = radial direction. $\rightarrow$ called major component tangential.

with high magnitude.

$P_y \Rightarrow$ responsible for dimensional inaccuracy & vibrations.



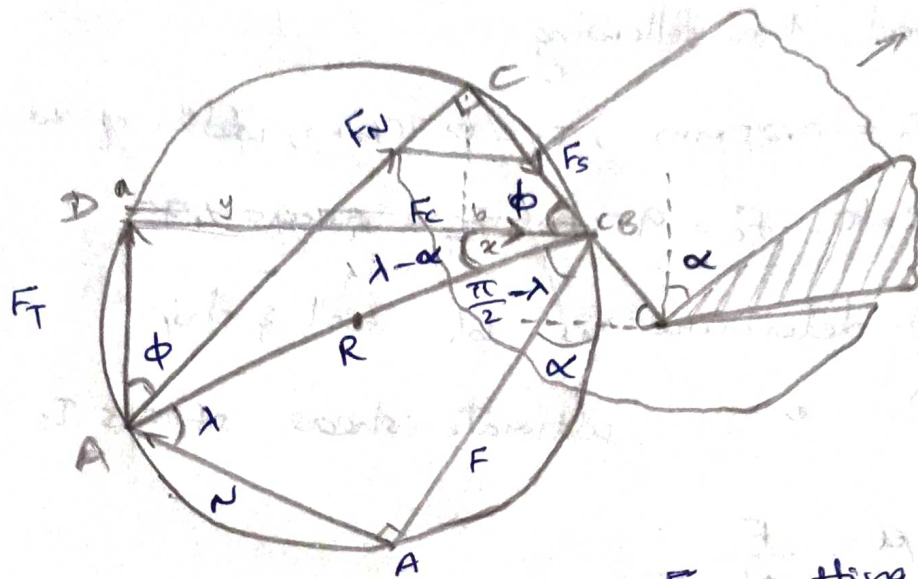
Development of Merchant's circle.



$\beta \Rightarrow \phi$
 $\gamma \Rightarrow \alpha$
 $\lambda \Rightarrow \lambda$

Merchant's circle diagram. with cutting force.

1-10-24.



$F_c \Rightarrow$ cutting component

$F_T \Rightarrow$ Thrust component

Rake, shear, friction angles are needed for MCD.

$$F_c = F_s \cos \phi + F_N \sin \phi$$

$$F_s = F_c - F_T$$

$$F_N = F_c + F_T$$

$$F_T = F_N - F_s$$

$$F = F_c \sin \alpha + F_T \cos \alpha$$

$$N = F_c \cos \alpha - F_T \sin \alpha$$

$$\left(\mu = \frac{F}{N} \right)$$

$$R = \frac{F_s}{\cos (\phi + \lambda - \alpha)}$$

$$F_c = R \cos (\lambda - \alpha)$$

$$F_T = R \sin (\lambda - \alpha)$$

$$A_s = \frac{t_1 w}{\sin \phi}$$

$$\tan \phi = \frac{\alpha \cos \alpha}{1 - \alpha \sin \alpha}$$

$$= \alpha = \frac{t_1}{t_3} = \frac{1}{3}$$

$$\therefore \phi = \alpha$$

1. During an orthogonal machining operation on mild steel, the following

$T_1 = 0.25 \text{ mm}$, $T_2 = 0.75 \text{ mm}$, width of $w = 0.25 \text{ mm}$

$\alpha = 0$, $F_c = 950 \text{ N}$, $F_T = ~~445~~ 475$

i). determine μ b/w tool & chip.

ii). " ultimate shear stress τ_s of work material

Ans.

$$\mu = \frac{F}{N}$$

$$F = F_c \sin \alpha + F_T \cos \alpha.$$

$$= 950 \sin 0 + 475 \cos 0.$$

$$= \underline{475 \text{ N}}$$

$$N = F_c \cos \alpha - F_T \sin \alpha$$

$$= 950 \cos 0 - 475 \sin 0$$

$$= 950$$

$$\therefore \mu = \frac{475}{950} = \underline{0.5}$$

$$\text{ii). } A_s = \frac{t_1 w}{\sin \phi} \Rightarrow \text{shear area.}$$

$$= r = \frac{t_1}{t_2} = \frac{1}{3}$$

$$\phi = \tan^{-1} \left(\frac{1}{3} \right) = 18.4^\circ$$

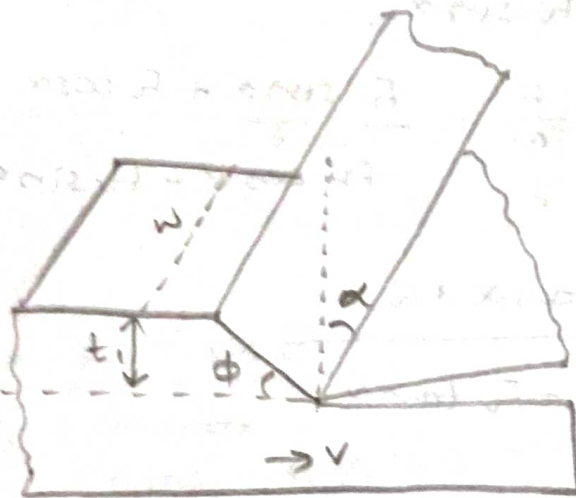
$$\therefore A_s = \frac{t_1 w}{\sin \phi} = 1.98 \text{ mm}^2$$

$$F_s = F_c \cos \phi - F_T \sin \phi$$

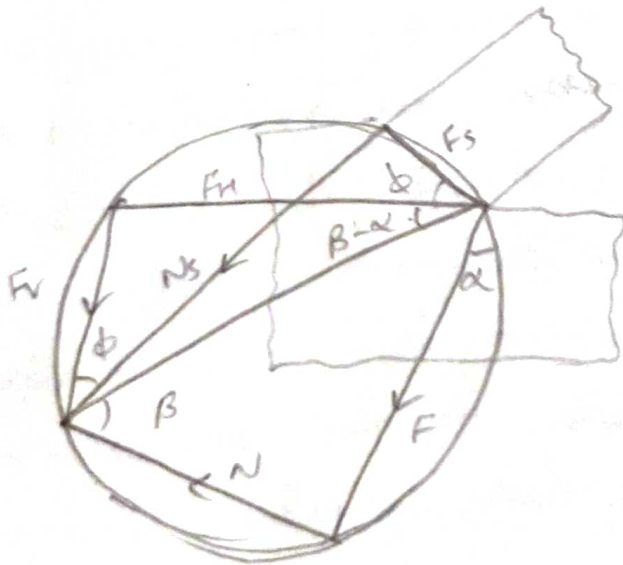
$$= 950 \cos(18.4) - 475 \sin(18.4)$$

$$= 751.3 \text{ N}$$

$$\tau_s = \frac{F_s}{A_s} = \frac{751.3}{1.98} = \underline{\underline{379.4 \text{ N/mm}^2}}$$



Merchant's force analysis.



Merchant's circle diagram makes it easy to determine the cutting force relations in metals.

The diameter of the circle & the resultant force is denoted by F .

$$F = F_H \sin \alpha + F_V \cos \alpha.$$

$$N = F_H \cos \alpha - F_V \sin \alpha.$$

$$\mu = \tan \beta = \frac{F}{N} = \frac{F_H \sin \alpha + F_V \cos \alpha}{F_H \cos \alpha - F_V \sin \alpha}$$

$$= \frac{F_H \tan \alpha + F_V}{F_H - F_V \tan \alpha}.$$

$$F_s = F_H \cos \phi - F_V \sin \phi$$

$$N_s = F_V \cos \phi + F_H \sin \phi = F_s \tan (\phi + \beta - \alpha)$$

t = unit chip thickness,

b = width of cut.

$$A_s = \frac{t}{\sin \theta} \times b.$$

τ = shear stress,

σ = Normal stress.

$$F_H = F_s \cos \phi + N_s \sin \phi$$

$$F_V = N_s \cos \phi - F_s \sin \phi$$

$$F_H = F_s \cos \phi + F_s [\tan (\phi \tan \beta - \alpha) \sin \phi]$$

$$F_H = F_s [\cos \phi + \tan (\phi + \beta - \alpha) \sin \phi]$$

$$= F_s \left[\cos \phi + \frac{\sin (\phi + \beta - \alpha)}{\cos (\phi + \beta - \alpha)} \cdot \sin \phi \right]$$

$$= F_s \frac{[\cos \phi \cos (\phi + \beta - \alpha) + \sin (\phi + \beta - \alpha) \sin \phi]}{\cos (\phi + \beta - \alpha)}$$

$$F_H = F_s \cdot \frac{\cos (\beta - \alpha)}{\cos (\phi + \beta - \alpha)} = \frac{\tau b t (\cos (\beta - \alpha))}{\sin \phi \cdot \cos (\phi + \beta - \alpha)}$$

$$F_v = \frac{\tau b t \sin (\beta - \alpha)}{\sin \phi \cdot \cos (\phi + \beta - \alpha)}$$

8-10-24.
1. In orthogonal turning, of a cylindrical tube of wall thickness 5mm, axial & tangential cutting forces were measured at 1259 N & 1601 N respectively. The measured chip thickness, after machining was found to be 0.3mm. Rake angle was 10° & the axial feed was 100mm/min. Rotational speed of spindle was 1000rpm. Find out shear strength of the material, assuming that the ^{material} pipe is perfectly plastic.

Ans. wall thickness, $t = 5\text{mm}$,

$F_a = 1259\text{N}$, $F_t = 1601\text{N}$. chip thickness, $t_c = 0.3\text{mm}$

Rake angle, $\alpha = 10^\circ$. Axial feed = 100mm/min $\Rightarrow f_v$

$$\text{Rotational speed} = 1000 \text{ rpm} = N.$$

$$\tau = \frac{F_s}{A_s}$$

$$F_s = F_c \cos \phi - F_t \sin \phi.$$

$$F_c = F_s \cos \phi + F_t \sin \phi.$$

For orthogonal cutting, $\lambda = 90^\circ$.

$$F_c = F_t = \underline{1601 \text{ N}}, \quad b = \frac{d}{\sin \lambda} = \frac{5}{\sin 90} = \underline{5 \text{ mm}}$$

$$f = \frac{100}{1000} = 0.1 \text{ mm/rev.}$$

$$t = f \sin \lambda = 0.1 \sin 90 = \underline{0.1 \text{ mm.}}$$

$$\lambda = \frac{t}{t_c} = \frac{0.1}{0.3} = \underline{0.33}.$$

$$\tan \phi = \frac{\lambda \cos \alpha}{1 - \lambda \sin \alpha} = \frac{0.33 \cos 10^\circ}{1 - 0.33 \sin 10^\circ}$$

$$\Rightarrow \phi = \underline{19.02^\circ}.$$

$$\text{Shear strength, } \tau_s = \frac{F_s}{A_s} = \frac{F_s \sin \phi}{bt}.$$

$$\tau_s = \frac{(F_c \cos \phi - F_t \sin \phi) \sin \phi}{bt}$$

$$= \frac{(1601 \cos 19.02 - 1259 \sin 19.02) \sin 19.02}{5 \times 0.1}$$

$$= \underline{719.12 \text{ MPa}}$$

10-10-24

Advantages of Merchant circle diagram (MCD)

Rake angle, shear angle & friction angle are needed for MCD.

F_c || to cutting velocity $\therefore$ it is responsible for work, power, energy.

Development of equation for estimation of cutting force

- Analytical method:
- Experimental method.

Tangential or main component;

$$F_c = P_z = R \cos(\eta - \gamma) = R \cos(\lambda - \alpha) \quad \text{--- (1)}$$

$$F_s = P_s = R \cos(\beta + \eta - \gamma) = R \cos(\phi + \lambda - \alpha) \quad \text{--- (2)}$$

dividing (1) by (2);

$$P_z = P_s \cos(\phi - \alpha) / \cos \phi - \lambda$$

$$P_z = t f \tau_s (\sec \gamma - \tan \gamma - 1)$$

For brittle work materials, like grey cast iron,

$$2\beta + \eta - \gamma = 90^\circ$$

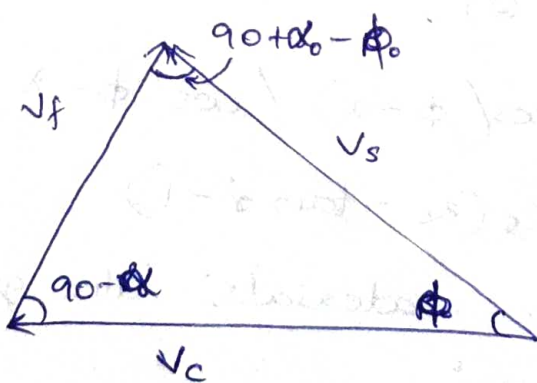
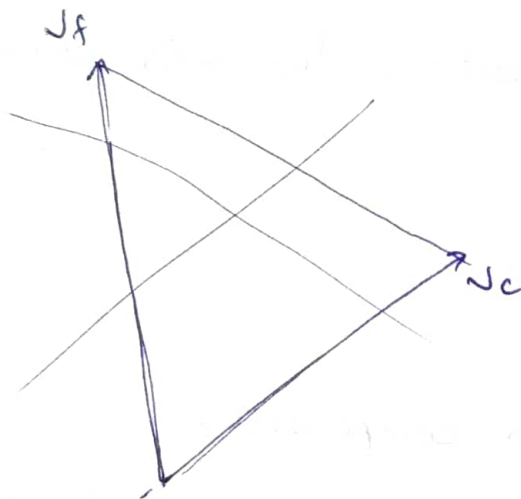
$$\phi + \lambda - \alpha = 45^\circ$$

$$P_{xy} = t f \tau_s (\sec \gamma - \tan \gamma - 1)$$

$$P_x = t f \tau_s (\sec \gamma - \tan \gamma - 1) \sin \phi$$

Metal cutting theories.

- Ernst.
- Modified Merchant circle.
- Lee and Shaffer's theory.



V_f = flow velocity of chip.

V_c = cutting velocity

V_s = shear velocity.

Metal removal rate : volume of metal removed in unit time.

$$MRR = t f V_c$$

t = depth of cut.

f = feed.

V_c = cutting speed.

With optimum MRR, cost of machining can be reduced.

Cutting power consumption, $P_c = P_z \cdot P_c + P_x \cdot V_f$.

V_f = feed velocity = $Nf/1000$ m/min

N = rpm.

$$P_c \approx P_z \cdot V_c$$

→ Cutting tools generally fail by:

- Mechanical failure due to excessive force & shocks.
- Quick dulling by plastic deformation.
- Gradual wear.

Crater & flank wear.

↓ ↓
Rake side clearance side